

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Eighth Semester of B. Tech (EE) Examination

Apr-May 2018

EE409 Digital Signal Processing

Date: 07.05.2018, Monday

Time: 01.30 p.m. To 04.30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q – 1. Give advantage of digital signal processing over analog signal processing and application of digital signal processing. **[06]**

Q – 2 Attempt any two. **[14]**

- (a)** Find inverse Z-Transform using partial fraction expansion method if

$$X(z) = \frac{1}{(1+z^{-1})(1-z^{-1})^2}$$

- (b)** Obtain the Cascade realization of the system characterized by the transfer function

$$H(z) = \frac{1+(1/3)Z^{-1}}{\left[1-(1/2)Z^{-1}+(1/3)Z^{-2}\right]\left[1-(1/3)Z^{-1}+(1/2)Z^{-2}\right]}$$

- (c)** List out any five properties of Z- transform and prove the same..

- (d)** Find the Even and Odd Component of the following signals.

(i). $x(n) = \{-3, 1, 2, -4, 2\}$ **(ii).** $x(n) = \{5, 4, 3, 2, 1\}$ **(iii).** $x(n) = \{-5, 2, 1, -3, 1\}$

Q – 3 Attempt any three. **[15]**

- (a)** Derive the transfer function of IIR system and draw the general realization structure for Direct Form - I of IIR System.

- (b)** Write any five properties of Discrete Time Fourier Transform (DTFT).

- (c)** Obtain the Direct Form-I from realization of the LTI system governed by the equation.

$$y(n) = -\frac{13}{12}y(n-1) - \frac{9}{24}y(n-2) - \frac{1}{24}y(n-3) + x(n) + 4x(n-1) + 3x(n-2)$$

- (d)** Define the following.

- (i). Causal and Non-Causal System
- (ii). Static and Dynamic System
- (iii). Region of Convergence (ROC)

SECTION – II

Q – 4. Write the procedure to design of FIR filters using windows. [06]

Q - 5 (a) Given $x(n) = \{1, 2, 3, 4, 4, 3, 2, 1\}$. Find $X(k)$ using DIF FFT algorithm. [07]

(b) Design a low-pass FIR filter with five stage [given: sampling time 1 ms; $f_c = 200$ Hz]. [07]
Also find the frequency response of the filter.

OR

(b) Given $x(n) = \{0, 1, 2, 3, 4, 5, 6, 7\}$. Find $X(k)$ using DIT FFT algorithm. [07]

Q – 6 Attempt any three. [15]

(a) Find the convolution of the following using Tabular Method.

(i). $x(n) = \{2, 3, 2, 2\}$, $h(n) = \{1, -2, 3, -1\}$

(ii). $x(n) = \{1, 2, -1, -2, 3, 1\}$, $h(n) = \{3, 2, 1\}$

(b) Find the circular convolution of $x_1 = \{1, 2, 1, 2\}$ and $x_2 = \{4, 3, 2, 1\}$.

(c) Compute and plot the convolution using graphical Method.

If $x(n) = \{4, 2, 1, 3\}$ and $h(n) = \{1, 2, 2, 1\}$

(d) For the analog transfer function $H(s)$ determine $H(z)$ when $T = 1$ s and using bilinear transformation method. Where $H(s) = \frac{3}{s^2 + 0.5s + 2}$

(e) Define linear and Non-linear System also check whether the given system is linear or not

$$Y(n) = x(n) + \frac{1}{2x(n-2)}$$
